Malware Analysis Report

WannaCry Ransomware Investigation

Classification: CONFIDENTIAL

Target: sandra.odwyer@redacted.com (VIP) | Alert Date: 11 March 2024

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VERDICT	MALWARE FAMILY	THREAT SCORE	VT DETECTION
MALICIOUS	WannaCry	100 / 100	68 / 72

1. EXECUTIVE REPORT

On March 11, 2024, at 09:14 UTC, a CrowdStrike CSP alert was activated on the Cisco Email Security Appliance (ESA) due to a suspicious inbound email aimed at sandra.odwyer@redacted.com, a VIP user. The email was sent from a UUID-based address at azxmarket.com, routed through the SendCloud bulk mailing platform from the source domain duchessgarden.sn (Senegal TLD), and included a RAR archive attachment named Catalogue.rar.

Upon analyzing the attachment, it was found to contain a Windows executable called diskpart.exe, which is a WannaCry ransomware binary disguised as the legitimate Windows disk partitioning tool. This file was identified as malicious by 68 out of 72 security vendors on VirusTotal and achieved a perfect threat score of 100/100 with 100% confidence on Joe's Sandbox. The Cisco ESA antivirus scanner provided a negative verdict, indicating that the email gateway failed to detect the payload, with CrowdStrike's behavioral detection being the sole control that recognized the threat.

WannaCry is a ransomware strain of global significance, known for the 2017 worldwide cyberattack that compromised over 200,000 systems in 150 countries, resulting in estimated damages of $4–8 billion. It takes advantage of the EternalBlue SMB vulnerability (MS17-010) for lateral movement and encrypts victim files with a .WNCRY extension, demanding Bitcoin for decryption. The targeting of a VIP user at a financial institution with this payload suggests a high-value, targeted attack.

2. ALERT DETAILS

Field	Detail
Alert Name	alert_all_security_Crowdstrike_csp_IAAT_A2
Alert Source	Crowdstrike.CSP — Cisco ESA (messaging_cisco_esa index)
Alert Time	11 March 2024, 09:14:54 UTC
Host	remacdmzma01
Email Recipient	sandra.odwyer@redacted.com (VIP)
Email Sender	c6d90d50-df87-11ee-a2fc-525400450766@azxmarket.com
Source Domain	https://duchessgarden.sn/ (Senegal TLD)
Routing Infrastructure	ucmail288.sendcloud.org (bulk mail platform)
Attachment Name	Catalogue.rar
File Name Inside RAR	diskpart.exe
File Signature (SHA-256)	ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa
ESA AV Verdict	NEGATIVE — email gateway failed to detect the payload
Assignment	SECOAN Security Operations Centre

3. INVESTIGATION STEPS

- Alert triage: Reviewed the CrowdStrike CSP alert from the Cisco ESA. Identified sender anomalies, UUID-based sender address, bulk mail routing infrastructure, Senegal TLD source domain inconsistent with claimed business identity.
- Static analysis: Submitted the file hash (SHA-256: ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa) to VirusTotal. Confirmed 68/72 vendor detections, WannaCry family label, and retrieved all three cryptographic hashes.
- Sender infrastructure investigation: Submitted azxmarket.com and duchessgarden.sn to VirusTotal for domain reputation analysis.
- Dynamic analysis: Located prior detonation analysis on Joe's Sandbox (Analysis ID: 1878098) matching the exact SHA-256 hash. Reviewed full behavioural report including process tree, dropped files, network activity, registry modifications, and MITRE ATT&CK mapping.

IOC extraction: Compiled all indicators of compromise from static and dynamic analysis for threat intelligence and Sentinel alert creation.

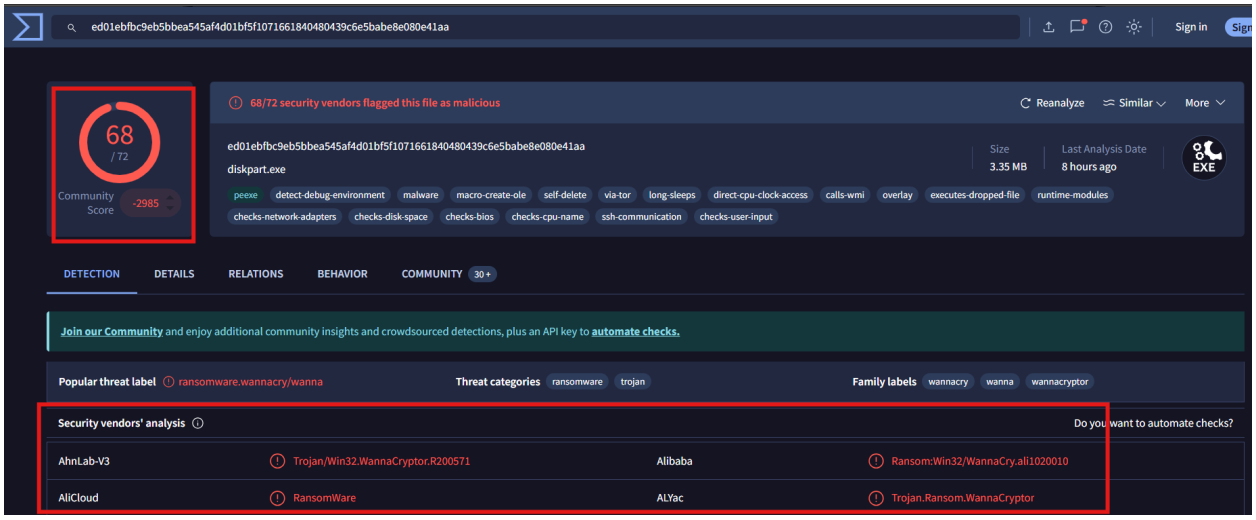
Incident response planning: Assessed containment, eradication, and recovery actions based on confirmed WannaCry behaviour.

4. FINDINGS & EVIDENCE

4.1 File Identification & Signatures

The file delivered inside Catalogue.rar is named diskpart.exe, chosen to impersonate the legitimate Windows disk partitioning utility (C:\Windows\System32\diskpart.exe). This is a deliberate masquerading technique (MITRE T1036) designed to reduce suspicion if the filename is observed by a user or analyst.

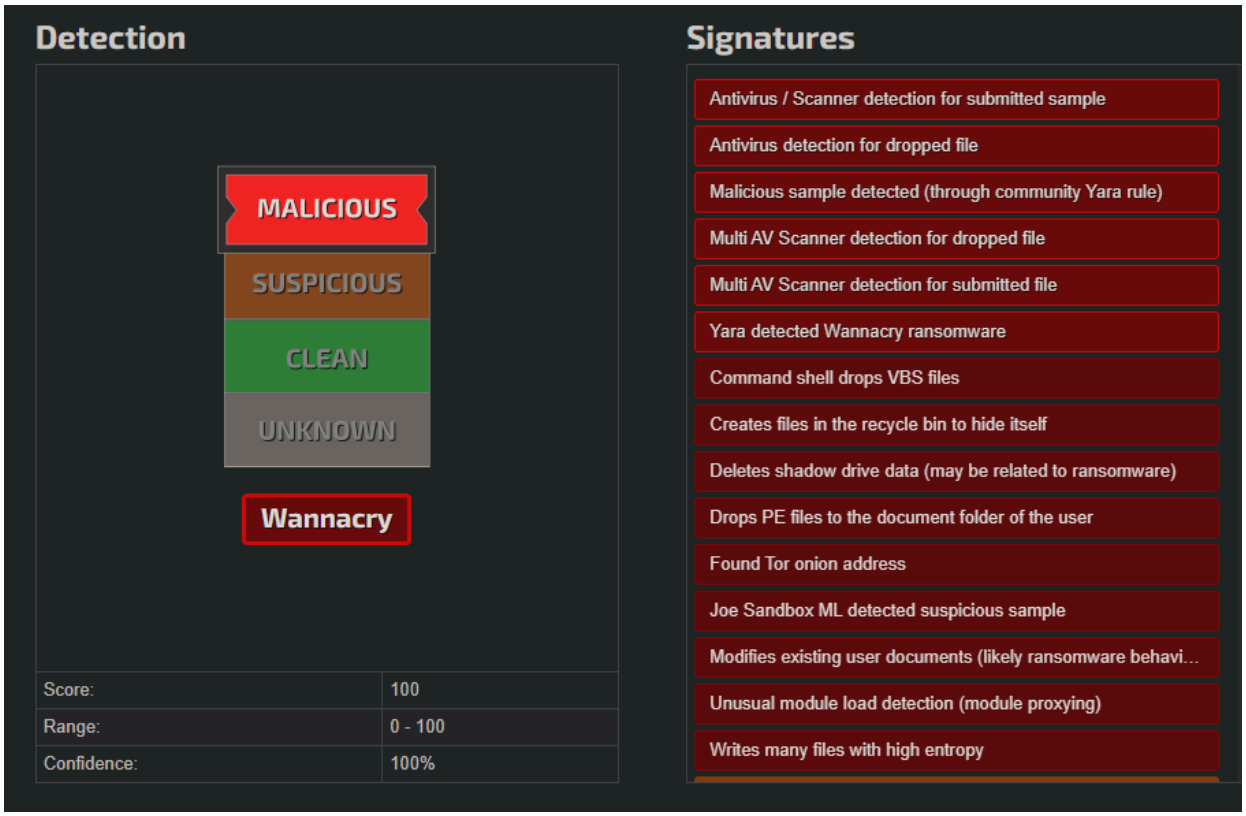
Property	Value
File Name	diskpart.exe (delivered inside Catalogue.rar)
File Type	Win32 EXE - PE32 executable (GUI) Intel 80386, MS Windows
File Size	3.35 MB (3,514,368 bytes)
Compiler	Microsoft Visual C++ (PE32, 32-bit)
MD5	84c82835a5d21bbcf75a61706d8ab549
SHA-1	5ff465afaabcbf0150d1a3ab2c2e74f3a4426467
SHA-256	ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa
Signature Type	SHA-256 (primary IOC)
VirusTotal Detection	68 / 72 vendors - Popular label: ransomware.wannacry/wanna
Family Labels	wannacry, wanna, wannacryptor
Joe's Sandbox Score	100 / 100 — Confidence: 100% — Verdict: MALICIOUS
Ransomware Name	WannaCry (also known as WannaCryptor, WCry, Wanna Decryptor)



VirusTotal — 68/72 detection, WannaCry family label, ransomware/trojan threat categories

Basic properties	
MD5	84c82835a5d21bbcf75a61706d8ab549
SHA-1	5ff465afaabcbf0150d1a3ab2c2e74f3a4426467
SHA-256	ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa
Vhash	036046656d1570a8z3631lz1fz
Authentihash	4b2c4c7f06f5ffaeea6efc537f0aa66b0a30c7ccd7979c86c7f4f996002b99fd
Imphash	68f013d7437aa653a8a98a05807afeb1
Rich PE header hash	417a06d07f984f3bce5cd06546c98842
SSDEEP	98304:QqPoBhz1aRxcSUDk36SAEdhvxWa9P593R8yAVp2g3x:QqPe1Cxcxk3ZAEUadzR8yc4gB
TLSH	T173F533F4E221B7ACF2550EF64855C59B6A9724B2EBEF1E26DA8001A70D44F7F8FC0491
File type	Win32 EXE executable windows win32 pe peexe
Magic	PE32 executable (GUI) Intel 80386, for MS Windows
TrID	Win32 Executable MS Visual C++ (generic) (39.7%) | Microsoft Visual C++ compiled executable (generic) (21%) | Win32 Dynamic Link Library (generic) (8.3%) | Win64 Executable (generic) (8.3%) ...
DetectItEasy	PE32 | Compiler: EP:Microsoft Visual C/C++ (6.0 (1720-9782)) [EXE32] | Compiler: Microsoft Visual C/C++ (12.00.9782) [C++] | Linker: Microsoft Linker (6.00.8047) | Tool: Visual Studio (6.0) | Ar...
Magika	PEBIN
File size	3.35 MB (3514368 bytes)
PEiD packer	Microsoft Visual C++

VirusTotal Details tab — MD5, SHA-1, SHA-256, file type, size, compiler.



Joe's Sandbox — Detection panel showing Score 100/100, Confidence 100%, Verdict MALICIOUS, WannaCry family label, full signatures list

4.2 Key Signature Analysis

The key signature associated with this alert is the SHA-256 hash ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa. This hash has been confirmed across multiple threat intelligence platforms as a known WannaCry binary. The Joe's Sandbox behavioural analysis confirms the following key signatures triggered during execution:

- Yara detected WannaCry ransomware — YARA rule match by Florian Roth, definitive identification

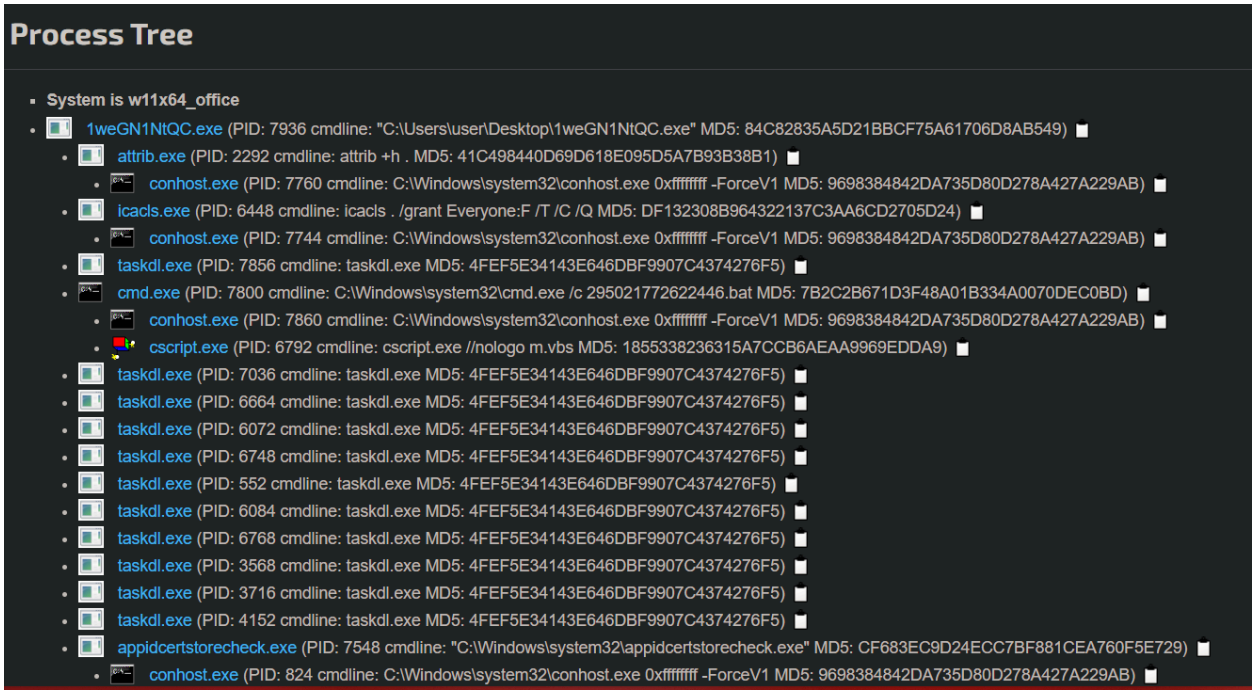
- Deletes shadow drive data — Volume Shadow Copy deletion prevents file recovery via Windows backup
- Modifies existing user documents — confirms file encryption behaviour
- Writes many files with high entropy — encrypted files produce high-entropy output, signature of ransomware encryption
- Found Tor onion address — C2 payment infrastructure embedded in binary
- Command shell drops VBS files — scripting for lateral movement
- Creates files in the recycle bin to hide itself — evasion technique
- Drops PE files to the document folder — executable payload distribution
- Unusual module load detection (module proxying) — DLL side-loading for defence evasion

4.3 Process Execution & Behaviour

Dynamic analysis on Joe's Sandbox (Windows 11 x64 with Office environment) confirmed full WannaCry execution. The process tree reveals the complete attack chain:

Process	MD5	Purpose
diskpart.exe (PID 7936)	84c82835a5d21bbcf75a61706d8ab549	Primary WannaCry binary - entry point
attrib.exe (PID 2292)	41C498440D69D618E095D5A7B93B38B1	Sets +h attribute to hide malicious files
icacls.exe (PID 6448)	DF132308B964322137C3AA6CD2705D24	Grants Everyone full permissions - icacls . /grant Everyone:F /T /C /Q
taskdl.exe (×hundreds)	4FEF5E34143E646DBF9907C4374276F5	WannaCry encryption worker - spawns hundreds of instances for parallel file encryption
cmd.exe (PID 7800)	7B2C2B671D3F48A01B334A0070DEC0BD	Executes 295021772622446.bat - ransomware batch configuration script
cscript.exe (PID 6792)	1855338236315A7CCB6AEAA9969EDDA9	Executes m.vbs - VBScript lateral movement component
@WanaDecryptor@.exe	N/A (dropped)	Ransom note display application - shown to victim

The mass spawning of taskdl.exe across hundreds of parallel processes is WannaCry's characteristic threading behaviour, designed to maximise encryption speed across all directories before the victim or security team can respond.



Joe's Sandbox Process Tree — top section showing diskpart.exe → attrib.exe, icacls.exe, taskdl.exe, cmd.exe → cscript.exe execution chain

4.4 Dropped Files

WannaCry dropped the following key files across user directories during execution:

- C:\@Please_Read_Me@.txt - WannaCry ransom note (YARA rule WannaCry_RansomNote matched). Contains: 'Q: What's wrong with my files? A: Don't worry about decryption.'
- C:\@WanaDecryptor@.exe - Ransom UI application dropped in root C: and all user directories (Desktop, Documents, Downloads, AppData). Detected 97–100% by Avira and ReversingLabs as Win32.Ransomware.WannaCry
- C:\Users\user\Desktop\taskdl.exe - Encryption worker binary (96% detection)
- C:\Users\user\Desktop\taskse.exe - Task scheduler component (89% detection)
- C:\Users\user\Desktop\u.wnry - Ransom UI resource file
- C:\Users\user\Desktop\295021772622446.bat - Ransomware batch configuration script (YARA rule WannCry_BAT matched)
- C:\Users\user\Desktop\m.vbs - VBScript lateral movement component
- *.WNCRY files - encrypted victim files with WannaCry extension appended

JoeSandbox Cloud BASIC

Overview Signatures Domains / IPs Dropped Static Network Stats Behavior Disa

Dropped Files

Source	Detection	Scanner	Label	L
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	100%	Avira	TR/FileCoder.724645	
C:\@WanaDecryptor@.exe	100%	Avira	LNK/Runner.VPDJ	
C:\@WanaDecryptor@.exe	97%	ReversingLabs	Win32.Ransomware.WannaCry	
C:\Users\user\AppData\Local\@WanaDecryptor@.exe	97%	ReversingLabs	Win32.Ransomware.WannaCry	

Joe's Sandbox Dropped Files — showing @WanaDecryptor@.exe paths, taskdl.exe, taskse.exe, u.wnry with detection percentages

4.5 Network Behaviour

WannaCry's network activity confirms two distinct behaviours: C2 communication via Tor and lateral movement via SMB.

Kill Switch Domain Check

Before activating, WannaCry performs a DNS lookup for the kill switch domain: www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com. If this domain resolves (as it did after security researcher Marcus Hutchins registered it in May 2017), WannaCry terminates. In the sandbox environment, the domain did not resolve and execution proceeded.

Tor C2 Infrastructure

The following Tor onion addresses were found embedded in both the main binary and the c.wnry resource file, these are WannaCry's payment and C2 servers:

C13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94gx7ekbenv2riucmf.onion

57g7spgrzlojinas.onion

xxlvbrloxvriy2c5.onion

76jdd2ir2embyv47.onion

cwwnhwhlz52maqm7.onion

WannaCry also attempts to download Tor browser if not present on the system via: https://dist.torproject.org/torbrowser/6.5.1/tor-win32-0.2.9.10.zip

SMB Lateral Movement

WannaCry scans the local network for hosts with port 445 (SMB) open and exploits the EternalBlue vulnerability (MS17-010) on unpatched Windows systems to propagate without user interaction. The sandbox captured 566 IP connections consistent with subnet-range SMB scanning.

Networking

Found Tor onion address	
Source: 1weGN1NtQC.exe, 00000000.00000002.3002295255.000000001000C000.00000004.00001000.00020000.00000000.sdmp	String found in binary or memory: C13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94gx7ekbenv2riucmf.onion;57g7spgrzlojinas.onion;xxlvbrloxvriy2c5.onion;76jdd2ir2embyv47.onion;cwwnhwhlz52maqm7.onion;https://dist.torproject.org/torbrowser/6.5.1/tor-win32-0.2.9.10.zipl0v
Source: c.wnry.0.dr	String found in binary or memory: C13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94gx7ekbenv2riucmf.onion;57g7spgrzlojinas.onion;xxlvbrloxvriy2c5.onion;76jdd2ir2embyv47.onion;cwwnhwhlz52maqm7.onion;https://dist.torproject.org/torbrowser/6.5.1/tor-win32-0.2.9.10.zip

Joe's Sandbox Networking — Tor onion addresses found in binary and c.wnry resource file

4.6 Persistence Mechanisms

WannaCry establishes persistence through the following mechanisms, confirmed by Joe's Sandbox Sigma rule detections:

- Startup Folder File Write (Sigma: HIGH) - WannaCry wrote ~SDCE9B.tmp to C:\Documents and Settings\All Users\...\Microsoft\Windows\Start Menu\Programs\Startup\. Any file in the Startup folder executes automatically on every Windows login.
- Registry Run Keys (MITRE T1547.001) - Modifies HKLM\SOFTWARE\WannaCrypt and Windows service registry entries to maintain presence across reboots.
- Windows Service Installation - Creates a system service entry for persistence, allowing WannaCry to survive system restarts.
- VBScript execution via Cscript (Sigma: HIGH) - cscript.exe //nologo m.vbs executed from the ransomware batch script for scripted persistence and propagation.

Sigma Signatures

System Summary

Sigma detected: Startup Folder File Write	
Source: File created	Author: Roberto Rodriguez (Cyb3rWard0g), OTR (Open Threat Research): Data: EventID: 11, Image: C:\Users\user\Desktop\1weGN1NtQC.exe, ProcessId: 7936, TargetFilename: C:\Documents and Settings\All Users\Application Data\Application Data\Application Data\Application Data\Application Data\Application Data\Application Data\Application Data\Application Data\Microsoft\Windows\Start Menu\Programs\Startup\~SDCE9B.tmp

Sigma detected: WSF/JSE/JS/VBA/VBE File Execution Via Cscript/Wscript	
Source: Process started	Author: Michael Haag: Data: Command: cscript.exe //nologo m.vbs, CommandLine: cscript.exe //nologo m.vbs, CommandLine|base64offset|contains: (, Image: C:\Windows\SysWOW64\cscript.exe, NewProcessName: C:\Windows\SysWOW64\cscript.exe, OriginalFileName: C:\Windows\SysWOW64\cscript.exe, ParentCommandLine: C:\Windows\system32\cmd.exe /c 295021772622446.bat, ParentImage: C:\Windows\SysWOW64\cmd.exe, ParentProcessId: 7800, ParentProcessName: cmd.exe, ProcessCommandLine: cscript.exe //nologo m.vbs, ProcessId: 6792, ProcessName: cscript.exe

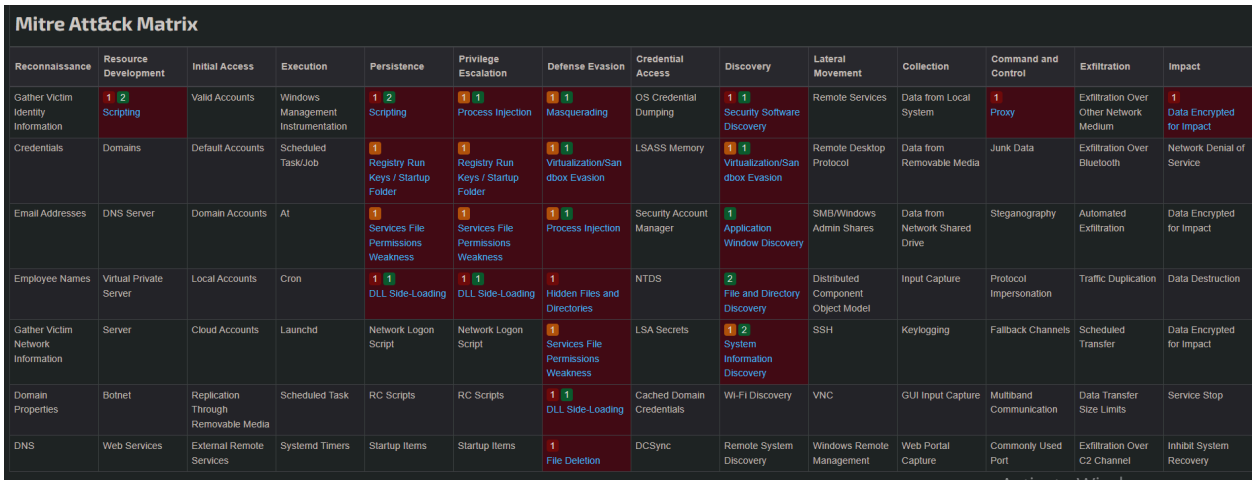
Joe's Sandbox Sigma Signatures — Startup Folder File Write and WSF/JSE/VBA/VBE File Execution Via Cscript/Wscript detections

4.7 MITRE ATT&CK Mapping

The following ATT&CK techniques were confirmed during behavioural analysis:

Tactic	Technique	Detail
Initial Access	**T1566.001 — Spearphishing Attachment**	WannaCry delivered via Catalogue.rar email attachment targeting a VIP user
Execution	**T1059.005 — VBScript**	cscript.exe executes m.vbs for lateral movement and propagation
Execution	**T1047 — WMI**	Windows Management Instrumentation used for system queries
Persistence	**T1547.001 — Registry Run Keys / Startup Folder**	Startup folder file write and registry modification for boot persistence
Persistence	**T1543 — Services File Permissions Weakness**	Windows service creation for persistent execution
Privilege Escalation	**T1055 — Process Injection**	Injects into legitimate system processes
Defense Evasion	**T1036 — Masquerading**	Malicious binary named diskpart.exe to impersonate legitimate Windows tool
Defense Evasion	**T1497 — Virtualization/Sandbox Evasion**	Detects debug environments; uses long sleep delays to evade sandbox timeouts
Defense Evasion	**T1574.002 — DLL Side-Loading**	Loads malicious DLLs via legitimate process DLL search order hijacking
Defense Evasion	**T1070.004 — File Deletion**	Deletes itself and shadow copies to hinder forensic recovery
Discovery	**T1082 — System Information Discovery**	Checks CPU name, BIOS, disk space, network adapters before activating
Discovery	**T1518 — Security Software Discovery**	Identifies installed security products to adapt evasion
Lateral Movement	**T1021 — Remote Services (SMB/EternalBlue)**	Exploits MS17-010 to propagate to unpatched hosts on port 445
Command & Control	**T1090 — Proxy (Tor)**	Uses Tor onion addresses for encrypted C2 and ransom payment
Impact	**T1486 — Data Encrypted for Impact**	Encrypts victim files with .WNCRY extension and demands Bitcoin ransom

Impact	**T1490 — Inhibit System Recovery**	Deletes Volume Shadow Copies to prevent recovery without paying ransom



Mitre Att&ck Matrix

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
Gather Victim Identity Information	1 2 Scripting	Valid Accounts	Windows Management Instrumentation	1 2 Scripting	1 1 Process Injection	1 1 Masquerading	OS Credential Dumping	1 1 Security Software Discovery	Remote Services	Data from Local System	1 Proxy	Exfiltration Over Other Network Medium	1 Data Encrypted for Impact
Credentials	Domains	Default Accounts	Scheduled Task/Job	1 Registry Run Keys / Startup Folder	1 Registry Run Keys / Startup Folder	1 1 Virtualization/Sandbox Evasion	LSASS Memory	1 1 Virtualization/Sandbox Evasion	Remote Desktop Protocol	Data from Removable Media	Junk Data	Exfiltration Over Bluetooth	Network Denial of Service
Email Addresses	DNS Server	Domain Accounts	At	1 Services File Permissions Weakness	1 Services File Permissions Weakness	1 1 Process Injection	Security Account Manager	1 Application Window Discovery	SMB/Windows Admin Shares	Data from Network Shared Drive	Steganography	Automated Exfiltration	Data Encrypted for Impact
Employee Names	Virtual Private Server	Local Accounts	Cron	1 1 DLL Side-Loading	1 1 DLL Side-Loading	1 Hidden Files and Directories	NTDS	2 File and Directory Discovery	Distributed Component Object Model	Input Capture	Protocol Impersonation	Traffic Duplication	Data Destruction
Gather Victim Network Information	Server	Cloud Accounts	Launchd	Network Logon Script	Network Logon Script	1 Services File Permissions Weakness	LSA Secrets	1 2 System Information Discovery	SSH	Keylogging	Fallback Channels	Scheduled Transfer	Data Encrypted for Impact
Domain Properties	Botnet	Replication Through Removable Media	Scheduled Task	RC Scripts	RC Scripts	1 1 DLL Side-Loading	Cached Domain Credentials	Wi-Fi Discovery	VNC	GUI Input Capture	Multiband Communication	Data Transfer Size Limits	Service Stop
DNS	Web Services	External Remote Services	Systemd Timers	Startup Items	Startup Items	1 File Deletion	DCSync	Remote System Discovery	Windows Remote Management	Web Portal Capture	Commonly Used Port	Exfiltration Over C2 Channel	Inhibit System Recovery

Joe's Sandbox MITRE ATT&CK Matrix — full technique mapping

5. INDICATORS OF COMPROMISE

5.1 File IOCs

Type	Value	Description
SHA-256	ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa	diskpart.exe - primary WannaCry binary
MD5	84c82835a5d21bbcf75a61706d8ab549	diskpart.exe
SHA-1	5ff465afaabcbf0150d1a3ab2c2e74f3a4426467	diskpart.exe
MD5	4FEF5E34143E646DBF9907C4374276F5	taskdl.exe - WannaCry encryption worker
Filename	@WanaDecryptor@.exe	Ransom UI - dropped in all user directories
Filename	@Please_Read_Me@.txt	WannaCry ransom note
Filename	295021772622446.bat	Ransomware batch configuration script
Filename	m.vbs	VBScript lateral movement component
Filename	taskse.exe	Task scheduler component
Filename	u.wnry / c.wnry	WannaCry resource files
File Extension	*.WNCRY	Encrypted victim file extension

Registry Key	HKLM\SOFTWARE\WannaCrypt	Presence indicates active WannaCry infection
Startup File	~SDCE9B.tmp in Startup folder	WannaCry persistence mechanism

5.2 Network IOCs

Type	Value	Description
Email Sender	c6d90d50-df87-11ee-a2fc-525400450766@azxmarket.com	UUID-based automated sender - delivery infrastructure
Domain	duchessgarden.sn	Source domain - Senegal TLD, suspicious origin
Domain	azxmarket.com	Sender domain - bulk mail infrastructure
Tor Onion	C13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94gx7ekbenv2riucmf.onion	WannaCry C2 / payment server
Tor Onion	57g7spgrzlojinas.onion	WannaCry C2 / payment server
Tor Onion	xxlvbrloxvriy2c5.onion	WannaCry C2 / payment server
Tor Onion	76jdd2ir2embyv47.onion	WannaCry C2 / payment server
Tor Onion	cwwnhwhlz52maqm7.onion	WannaCry C2 / payment server
URL	https://dist.torproject.org/torbrowser/6.5.1/tor-win32-0.2.9.10.zip	Tor browser download - WannaCry installs Tor if absent
Domain	www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com	WannaCry kill switch domain - block resolution to prevent activation
Port	445 / SMB	WannaCry lateral movement - EternalBlue (MS17-010)

6. INCIDENCE RESPONSE & REMIDIATION

6.1 Immediate Containment

- Confirm whether sandra.odwyer@redacted.com opened the attachment or extracted diskpart.exe from Catalogue.rar. Check endpoint logs for execution of diskpart.exe from a non-system path (any path other than C:\Windows\System32\).
- If execution is confirmed: immediately isolate the affected device from the network, disconnect from all wired and wireless connections. Do not shut down, live memory may contain encryption keys.

- Block the following at email gateway: azxmarket.com, duchessgarden.sn, the UUID sender address pattern (*@azxmarket.com).
- Block all five Tor onion addresses and the kill switch domain at DNS level and firewall. Note: blocking the kill switch domain prevents WannaCry from self-terminating on new infections, consider allowing resolution rather than blocking.
- Block port 445 (SMB) inbound/outbound at the network perimeter immediately to prevent lateral movement. Internal SMB should be audited and restricted to necessary paths only.

6.2 Eradication

- Patch MS17-010 (EternalBlue) on all Windows systems immediately, this is WannaCry's primary propagation mechanism. Microsoft released patch MS17-010 in March 2017. Any unpatched system is vulnerable to automatic WannaCry infection without user interaction.
- Scan all systems for the SHA-256 hash ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa and the presence of @WanaDecryptor@.exe, @Please_Read_Me@.txt, or *.WNCRY files.
- Check all systems for the registry key HKLM\SOFTWARE\WannaCrypt - presence indicates infection.
- Remove the startup folder persistence file (~SDCE9B.tmp) from all affected systems.
- If Volume Shadow Copies have been deleted on affected systems, recovery from offline backups is required - encrypted files cannot be decrypted without the private key.
- Do not pay the ransom - WannaCry's payment infrastructure was disrupted in 2017 and decryption keys are not reliably delivered even after payment.

6.3 Recovery

- Restore encrypted files from the most recent clean backup taken before the infection timestamp. Verify backup integrity before restoration.
- Rebuild any fully compromised systems from clean images rather than attempting in-place recovery.
- After restoration, verify all systems are patched to MS17-010 level before reconnecting to the network.
- Conduct a full network scan to confirm no remaining WannaCry persistence on any host before declaring eradication complete.

6.4 Long-Term Recommendations

- Enforce email attachment policies - executable files inside archive formats (RAR, ZIP) should be blocked or sandboxed at the email gateway, not passed to endpoint AV alone. The Cisco ESA missed this payload; a sandboxing layer (e.g. Cisco Threat Grid) would have caught it.
- Disable SMBv1 across the entire environment - WannaCry requires SMBv1 for EternalBlue exploitation. SMBv1 has been deprecated by Microsoft and should not be enabled on any system.
- Implement network segmentation - limit internal SMB connectivity to prevent lateral movement. WannaCry's propagation depends on being able to reach port 445 on other hosts.
- Ensure offline, immutable backups are maintained and tested regularly - the only reliable recovery from WannaCry without payment is clean backups.
- Add all IOCs from Section 5 to Microsoft Sentinel custom threat intelligence and create detection alerts for the file hashes, Tor onion addresses, and WannaCry registry keys.

7. COMMUNICATION

7.1 To Sandra O'Dwyer (Affected VIP User)
To: sandra.odwyer@redacted.com Subject: URGENT — Security Alert on Email Received Today

Hi Sandra, our Security Operations team has flagged a malicious email sent to your inbox today containing ransomware in an attachment named Catalogue.rar. Please do not open this email or any attachments within it. If you have already opened or extracted any files from this attachment, please contact the security team immediately and do not use your device until we have assessed it. Our team is investigating and will update you shortly.

7.2 To CISO / Management
To: Marshall D Teach (CISO)Subject: CRITICAL — WannaCry Ransomware Delivery Attempt Against VIP User

This is to notify you of a confirmed WannaCry ransomware delivery attempt against sandra.odwyer@redacted.com (VIP) on 11 March 2024. The payload was delivered via a RAR archive disguised as a business catalogue and was not detected by the Cisco ESA antivirus scanner - it was caught only by the CrowdStrike behavioural alert. We are investigating whether the attachment was opened. If executed, this ransomware encrypts all accessible files, deletes shadow copies, and propagates via unpatched SMB across the network. Immediate actions are in progress including email gateway blocks, SMB port restrictions, and patch status review. Full analysis report attached.